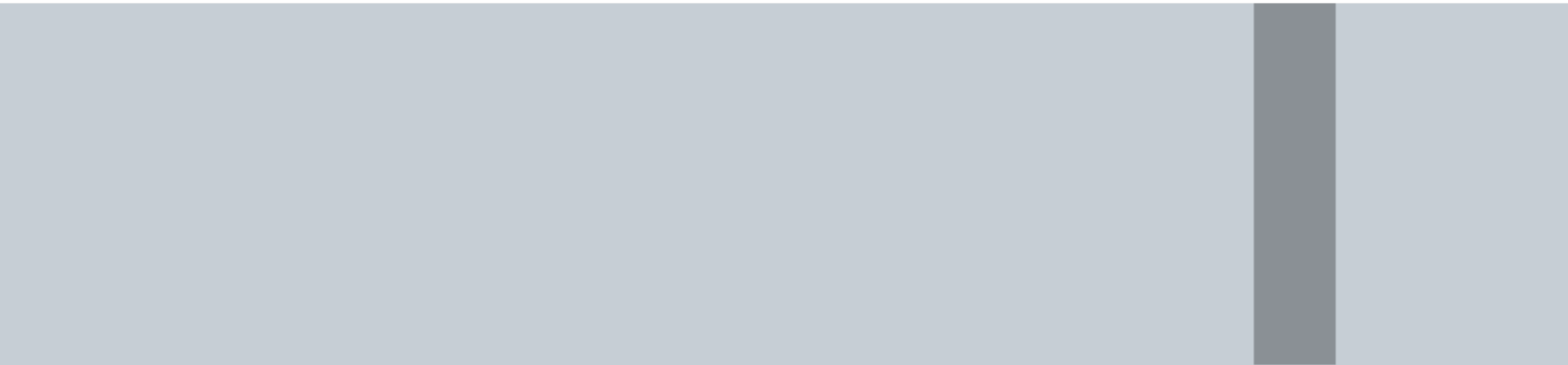
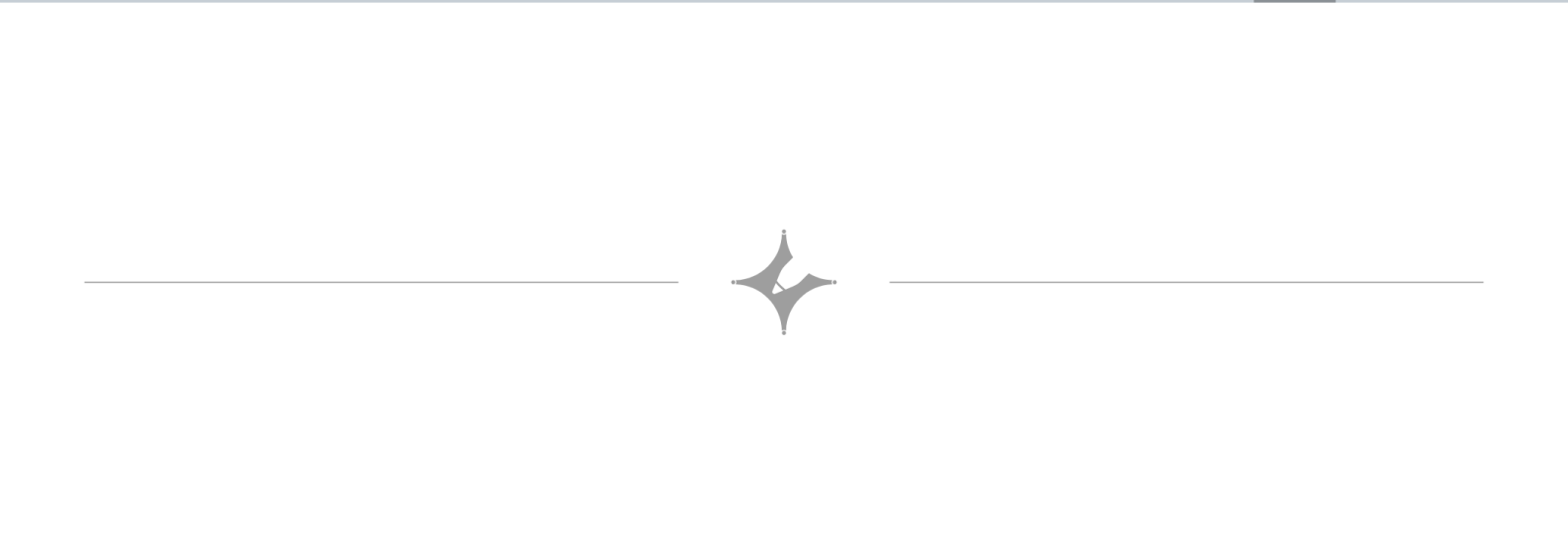
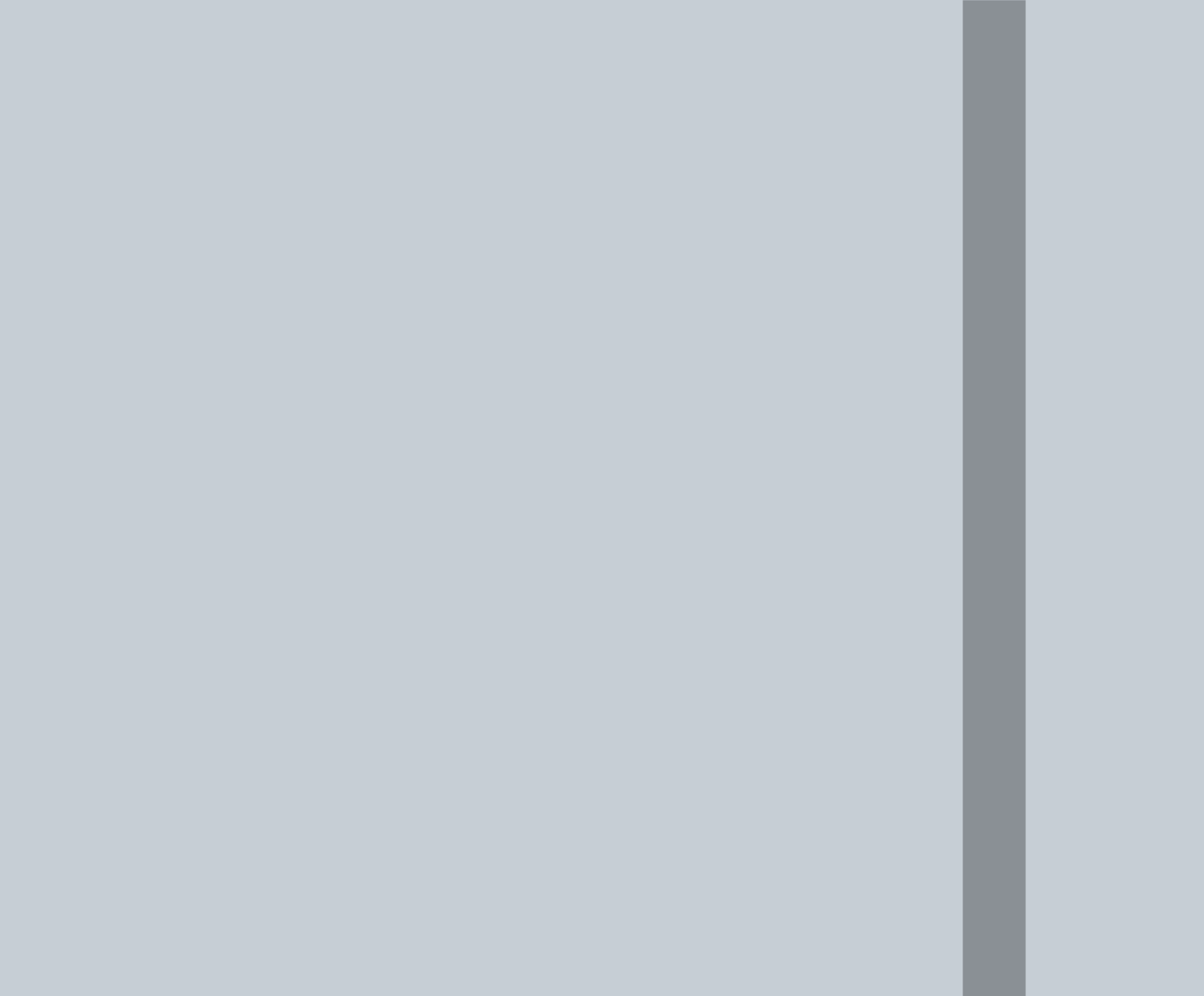




Logarithms

CONCEPT		
$\log_a(xy) = \log_a x + \log_a y$ $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$ $\log_x x = 1$ $\log_a 1 = 0$ $\log_a(x^n) = n(\log_a x)$ $\log_a x = \frac{1}{\log_x a}$ $\log_a x = \frac{\log_b x}{\log_b a} = \frac{\log x}{\log a}$	If $a^x = N$, Then, $\log_a N = x$ - Logarithms are defined only for positive numbers. - Logarithms are not defined for zero or negative numbers. Characteristic: The internal part of the logarithm. Mantissa: The decimal part of the logarithm. 12.3456 Natural log: Base "e". Common log: Base "10".	$a^m \times a^n = a^{m+n}$ $\frac{a^m}{a^n} = a^{m-n}$ $(a^m)^n = a^{mn}$ $(ab)^n = a^n b^n$ $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ $a^0 = 1$

3. The value of $\log 9/8 - \log 27/32 + \log 3/4$ is ?

- A. 0 B. 1 C. 2 D. 3

$$\log \left[\frac{9 \times 32 \times 3}{8 \times 27 \times 4} \right] = \log 1 = 0$$

4. The simplified form of $\log(75/16) - 2 \log(5/9) + \log(32/243)$?

- A. $2 \log 2$ B. $\log 2$ C. $\log 3$ D. $\log 5$

$$\log \left[\frac{75 \times 9^2 \times 32}{16 \times 5^2 \times 243} \right] = \log 2$$

6. The equation $\log_a(x) + \log_a(1+x) = 0$ can be written as?

- A. $x^2 + x + 1 = 0$ B. $x^2 + x - 1 = 0$
C. $x^2 - x - 1 = 0$ D. $x^2 - x + 1 = 0$

$$\begin{aligned} \log_a[x(1+x)] &= 0 \\ x(1+x) &= 1 \\ x + x^2 &= 1 \\ x + x^2 - 1 &= 0 \\ \underline{x^2 + x - 1} &= 0 \end{aligned}$$

$$\begin{aligned} \text{or } \log_a[x(1+x)] &= 0 \\ \log_a[x(1+x)] &= \log_a[1] \quad (\because \log_a 1 = 0) \\ \Rightarrow x^2 + x &= 1 \\ \Rightarrow \underline{x^2 + x - 1} &= 0 \end{aligned}$$

